# HIGH LEVEL DESIGN DOCUMENT

## Real-Time Luxury Auction Engine

Digital Auction & Bidding Platform

**Enterprise Architecture**
Confidential - Internal Project

## Real-Time Luxury Auction Engine

### Digital Auction & Bidding Platform

| Metadata | Value |
|---|---|
| **Prepared By :** | System Architecture & Delivery Team |
| **Date :** | 24 Oct 2026 |
| **Version :** | 1.0 |

## 1. Governance

### 1.1 Contributors

| Name | Position |
|---|---|
| Daniel H. S. | System Analyst / Solution Architect |
| Alini | Technical Lead |
| Geri Setyo Andrian | Project Manager |

### 1.2 Approvals

| Name | Position | Date | Signature |
|---|---|---|---|
| Geri Setyo Andrian | Project Manager Delivery Team | 24 Oct 2026 | `[ APPROVED & SIGNED ]` |
| Alini | Technical Architect / Tech Lead Delivery Team | 24 Oct 2026 | `[ APPROVED & SIGNED ]` |
| Daniel H. S. | System Analyst Delivery Team | 24 Oct 2026 | `[ APPROVED & SIGNED ]` |

### 1.3 Revision History

| Version | Date | Author | Summary of Changes |
|---|---|---|---|
| 0.1 | 24 Oct 2026 | Daniel H. S. | Initial Document Draft (Analysis & Discovery) |
| 0.5 | 24 Oct 2026 | Daniel H. S. | Added Dual-Model Auction, Velocity Anti-sniping, Redis Lua |
| 1.0 | 24 Oct 2026 | Daniel H. S. | Baseline Alignment with Enterprise Template Structure & Sign-Off |

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## 3. Introduction

### 3.1 High-Level Design Overview

Dokumen High-Level Design (HLD) ini menyajikan cetak biru arsitektur menyeluruh dari **Real-Time Luxury Auction Engine**. Platform ini mengkombinasikan pertukaran pesan real-time sub-300 ms, penahanan agunan finansial dinamis, pencegahan sniping berbasis lonjakan velositas penawaran, dan integritas ACID multi-tier.

### 3.2 Glossaries

| No | Terms | Expansion | Definition |
|---|---|---|---|
| 1 | **SSO** | Single Sign-On | Mekanisme otentikasi terpusat pihak ketiga melalui penyedia identitas NIRO. |
| 2 | **SLA** | Service Level Agreement | Standar performa latensi sistem (sinkronisasi data penawaran < 300 ms). |
| 3 | **BLOB** | Binary Large Object | Format penyimpanan data biner mentah gambar lot lelang pada kolom `BYTEA` PostgreSQL. |
| 4 | **ACID** | Atomicity, Consistency, Isolation, Durability | Kepatuhan transaksi database relasional pada mutasi saldo dan status aset. |
| 5 | **WSS** | WebSocket Secure | Protokol komunikasi dua arah dupleks penuh terenkripsi TLS untuk streaming data lelang. |
| 6 | **AM** | Auction Manager | Pengelola kurasi lot, pembuatan katalog barang mewah, dan monitoring event lelang. |

### 3.3 Objective Overview

Tujuan inisiatif ini adalah merancang dan membangun mesin lelang online berperforma tinggi yang mampu memproses beban lonjakan penawaran hingga 10.000 transaksi/detik tanpa terjadi race condition, menjamin kepastian dana agunan peserta lelang, dan memberikan transparansi real-time kepada seluruh penawar.

### 3.4 References

- **Product Requirement & Hackathon Brief 2026:** [PDF Brief] (file:///C:/Users/Daniel/.gemini/antigravity/brain/bd980ca4-23ee-44a3-a5ed-230fd0690b42/.user_uploaded/media_1792812187986_0083b4e8.pdf)
- **User Stories & Acceptance Criteria:** [ `user-stories-and-ac.md` ](file:///d:/Projects/Salt-Hackaton-2026/02_Projects/real-time-auction-engine/03_Requirements/user-stories-and-ac.md)
- **API Contracts Specification:** [ `api-contracts.md` ](file:///d:/Projects/Salt-Hackaton-2026/02_Projects/real-time-auction-engine/05_Integration/api-contracts.md)
- **Database Schema & Redis Lua Design:** [ `database-schema.md` ](file:///d:/Projects/Salt-Hackaton-2026/02_Projects/real-time-auction-engine/06_Data_Design/database-schema.md)

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## 4. Executive Summary

### 4.1 Purpose and Objectives

Sebagai platform lelang barang mewah modern, keandalan sistem bergantung pada pencegahan penipuan (*fraud*), likuiditas saldo riil saat menawar, serta keadilan waktu lelang. Inisiatif ini menghadirkan:

1. **Pencegahan Bot & Kecurangan:** Eliminasi input teks bebas melalui tombol inkremental diskrit yang telah ditentukan kurator.
2. **Dynamic Collateral Protection:** Dana peserta ditahan (*locked*) secara proporsional saat melakukan penawaran, dan dicairkan kembali secara otomatis 100% saat lelang berakhir (bagi yang kalah) atau didebit penuh (bagi pemenang).
3. **Pemberian Waktu Adil (Anti-Sniping Velocity):** Otomatis memperpanjang durasi lot bila terjadi lonjakan penawaran $\ge 10\text{ bid/s}$ dalam 2 menit terakhir penutupan lelang.

### 4.2 As-is vs To-be Architecture

| Karakteristik | Kondisi Umum (As-Is / Legacy) | Solusi Platform (To-Be Architecture) |
|---|---|---|
| **Mekanisme Penawaran** | HTTP Polling berulang / refresh manual halaman | Dupleks Penuh WebSocket (< 300 ms latency broadcast) |
| **Kunci Persaingan Saldo** | Table-lock / Row-lock PostgreSQL lambat | In-Memory Mutex Atomic via Redis Lua Scripting (< 5 ms) |
| **Validasi Agunan** | Saldo mengendap pasif kaku di awal pendaftaran | Penahanan dinamis per penawaran lelang yang aktif |
| **Penyimpanan Media** | Bergantung pada object storage cloud eksternal | Terkapsulasi aman di Postgres BLOB ( `BYTEA` ), disajikan dalam Base64 terkompresi |
| **Autentikasi** | Registrasi akun terisolasi manual | Single Sign-On (NIRO SSO) terpadu dengan auto-provisioning |

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## 5. Business Architecture

### 5.1 Actors & Roles

1. **Bidder (Peserta Lelang):** Pengguna terverifikasi NIRO SSO yang dapat melakukan top-up e-wallet, menelusuri katalog, melakukan penawaran inkremental tombol, dan mengaktifkan bot proxy.
2. **Auction Manager (Kurator / Admin Lelang):** Pengguna terverifikasi NIRO SSO yang dipromosikan untuk membuat listing lot lelang, mengunggah foto produk, menentukan harga dasar, kelipatan tombol bid, dan durasi anti-sniping.
3. **Superadmin:** Administrator utama platform yang mengelola hak akses promosi role, memantau audit trail, dan masuk melalui portal terisolasi email-password dengan pengamanan *progressive lockout*.

### 5.2 Business Capabilities

- **Katalog & Kurasi Lot Mewah:** Mendukung 5 kategori aset bergengsi (Luxury Watches, Designer Bags, Fine Jewelry, Collectible Vehicles, Rare Memorabilia).
- **Dual-Model Bidding:** Model Lelang Capped Buyout (pembelian instan saat menyentuh batas valuasi) dan Model Lelang Time-Window (penawar tertinggi di akhir periode).
- **Manajemen Dompet Agunan:** Ledger mutasi saldo riil tanpa biaya tersembunyi dengan batasan minimum top-up Rp 100.000.

### 5.3 Business Process Flow

```
flowchart LR
    A["Bidder Login via NIRO SSO"] ⟶ B["Top-Up Saldo E-Wallet"]
    B ⟶ C["Buka Live Room Lot Lelang"]
    C ⟶ D{"Pilih Tombol Inkrement"}
    D ⟶ E["Atomic Lua Validation"]
    E ⟶|Valid| F["Dana Ditahan (Locked) & Broadcast WS"]
    E ⟶|Saldo Kurang| G["Tolak Bid"]
    F ⟶ H{"Cek Kondisi Tutup"}
    H ⟶|Buyout Tercapai| I["Instant Settlement"]
    H ⟶|Waktu Habis| I
    I ⟶ J["Pemenang: Debit Saldo | Kalah: Release 100%"]
```

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## 6. Application & Integration Architecture

### 6.1 Integration Landscape

```
flowchart TD
    subgraph Presentation_Layer ["Presentation Tier (Next.js 14)"]
        FE_Bidder["Bidder Portal\n- Dynamic Countdown\n- Incremental Bid Controls\n- Real-time Ledger Balan
        FE_Admin["Admin & Superadmin Console\n- Lot Creation & Binary Upload\n- Live Auction Monitor Stream
    end

    subgraph Auth_Gateway ["Identity Provider"]
        NIRO["NIRO SSO Server\n(OAuth2 / OIDC Token Verification)"]
    end

    subgraph Core_Engine ["Application Tier (Golang High-Performance Monolith)"]
        APIGW["API Gateway / Transport Router\n(Rate Limiter, JWT Auth Guard)"]

        subgraph Domain_Services ["Domain Logic Modules"]
            AuthMod["Identity & RBAC Module"]
            LotMod["Auction Catalog & Lifecycle Module"]
            BidMod["Bid Execution & Engine Module"]
            WalletMod["Wallet & Ledger Module"]
            WSHub["WebSocket Event Hub Broadcaster"]
        end
    end

    subgraph Memory_Layer ["Hot-Path Caching (Redis 7)"]
        LuaEngine["Redis Lua Script Engine\n(bid_placement.lua)"]
        RedisState["In-Memory Hash & Expiry ZSET"]
        RedisPubSub["Pub/Sub Channel Bus"]
    end

    subgraph Storage_Layer ["Authoritative Storage (PostgreSQL 16)"]
        PostgresDB[("PostgreSQL Database\n- users & role mappings\n- auctions (BLOB image_data)\n- ledger_e
    end

    FE_Bidder ⟶|REST API| APIGW
    FE_Bidder ⟶|WebSocket /ws/v1/auctions| WSHub
    FE_Admin ⟶|REST API| APIGW
    FE_Bidder -·->|SSO Authentication| NIRO
    APIGW -·->|Verify Niro Token| NIRO

    APIGW ⟶ AuthMod
    APIGW ⟶ LotMod
    APIGW ⟶ BidMod
    APIGW ⟶ WalletMod

    BidMod ⟶|EVALSHA Atomic Execution| LuaEngine
    LuaEngine ⟶ RedisState
    LuaEngine ⟶|Publish Event| RedisPubSub
    RedisPubSub ⟶|Subscribe| WSHub

    LotMod ⟶ PostgresDB
    WalletMod ⟶ PostgresDB
    BidMod -·->|Persist Approved Bids| PostgresDB
```

### 6.2 Impacted Objects / Components

| No | Impact | Object Name | Functions |
|---|---|---|---|
| 1 | **New** | `Auth Service` | Menangani pertukaran token NIRO SSO, auto-registrasi bidder, dan autentikasi email Superadmin. |
| 2 | **New** | `Bid Engine (Lua)` | Eksekusi transaksi penawaran atomik < 5 ms pada Redis memory tier. |
| 3 | **New** | `Auction Catalog` | Pengelolaan data lot, upload gambar format biner multipart ke PostgreSQL BLOB, dan konversi output Base64 terkompresi. |
| 4 | **New** | `Settlement Daemon` | Background service yang memantau expiry lot melalui Redis ZSET dan mengeksekusi settlement ACID. |
| 5 | **New** | `WebSocket Hub` | Penyiaran mutasi harga dan timer ke ribuan klien secara real-time. |

### 6.3 High-Level Sequence Diagrams

Alur penawaran atomik berkecepatan tinggi dapat dilihat pada diagram berikut:

```
sequenceDiagram
    autonumber
    actor Bidder as Peserta Lelang
    participant UI as Next.js Client
    participant GW as API Gateway
    participant BE as Golang Bid Engine
    participant Redis as Redis 7 (In-Memory Lua)
    participant WS as WebSocket Hub
    participant DB as PostgreSQL 16

    Bidder->>UI: Klik Tombol Bid (+Rp 5.000.000)
    UI->>GW: POST /api/v1/auctions/{id}/bids
    GW->>BE: Forward Authed Request
    BE->>Redis: EVALSHA bid_placement.lua (user_id, lot_id, bid_amount)
    Note over Redis: Validasi status lot, saldo dompet,<br/>freeze saldo, catat highest bidder

    alt Bid Berhasil
        Redis-->>BE: Status OK {new_price, bidder_id, end_time}
        BE->>Redis: PUBLISH auction:{id}:events {price_update}
        Redis-->>WS: Message Event
        WS-->>UI: Broadcast ke Semua Klien (< 300 ms)
        BE-->>GW: 200 OK (Bid Accepted)
        GW-->>UI: Response Sukses
        BE-)DB: Asynchronous Persistence ke Tabel bids & wallets
    else Saldo Kurang / Outbid Kalah Cepat
        Redis-->>BE: Error (INSUFFICIENT_FUNDS / OUTBID)
        BE-->>GW: 400 Bad Request
        GW-->>UI: Tampilkan Notifikasi Error
    end
```

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## 7. Technology & Data Architecture

### 7.1 In-Memory Hot-Path Tier (Redis 7)

- **Struktur Lot Lelang (`auction:{lot_id}` - HASH):** Menyimpan status lelang, harga tertinggi terkini, ID penawar tertinggi, waktu berakhir, batas valuasi beli langsung, dan toleransi velositas penawaran.
- **Struktur Saldo Agunan (`wallet:{user_id}` - HASH):** Mengelola `available_balance` dan `locked_balance`.

- **Indeks Waktu Lelang ( `auctions:expiry_zset` - ZSET):** Score berupa Unix Timestamp dari `end_time` untuk mendeteksi penutupan lelang secara presisi per 1 detik.

### 7.2 Persistent Database (PostgreSQL 16)

- **Tabel Kunci:** `users`, `wallets`, `ledger_entries`, `auctions`, `bids`, `proxy_bids`, `asset_ownership`.
- **Integritas Gambar:** Kolom `image_data BYTEA` dan `image_mime_type VARCHAR` memastikan dokumen dan katalog lelang berdiri mandiri tanpa ketergantungan storage pihak ketiga.

### 7.3 Golang Clean Architecture Layers

Pemisahan kode mengikuti standar arsitektur bersih:

- `cmd/api`: Titik masuk aplikasi dan dependency injection.
- `internal/domain`: Entitas inti aturan lelang tanpa dependensi framework.
- `internal/usecase`: Orkestrasi alur bisnis lelang dan dompet.
- `internal/repository`: Implementasi driver PostgreSQL dan Redis Client.
- `internal/transport`: HTTP REST endpoints dan WebSocket Hub.

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## 8. Application Interfaces

Daftar antarmuka inti sistem terdiri dari 14 endpoint utama:

1. `GET /api/v1/auth/niro/authorize-url` — Dapatkan URL redirect SSO.
2. `POST /api/v1/auth/login/niro` — Tukar token SSO NIRO & provisioning otomatis.
3. `POST /api/v1/auth/login/email` — Login kredensial internal Superadmin.
4. `PATCH /api/v1/admin/users/:id/role` — Promosi wewenang user ke Auction Manager.
5. `GET /api/v1/auctions/active` — List katalog lelang aktif (termasuk pagination meta).
6. `GET /api/v1/auctions/:id` — Detail spesifik lot lelang (data Base64 gambar).
7. `POST /api/v1/auctions/:id/bids` — Pengiriman penawaran tombol inkremental.
8. `POST /api/v1/auctions/:id/proxy-bid` — Aktivasi bot penawar otomatis.
9. `DELETE /api/v1/auctions/:id/proxy-bid` — Pembatalan bot penawar otomatis.
10. `GET /api/v1/wallet/me` — Cek saldo riil dan saldo agunan tertahan.
11. `POST /api/v1/wallet/topup` — Top-up saldo khusus Bidder (min Rp 100.000).
12. `GET /api/v1/wallet/transactions` — Riwayat mutasi ledger finansial berhalaman.
13. `POST /api/v1/admin/auctions` — Listing lot lelang baru (multipart upload gambar BLOB & extend config).
14. `WS /api/v1/ws/auctions` — Koneksi duplex real-time telemetri lelang.

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## 9. Architectural Decision Records (ADR Highlights)

| Keputusan Arsitektur | Alternatif yang Dipertimbangkan | Solusi Terpilih & Rationale |
|---|---|---|
| **Penanganan Konkurensi Bidding** | PostgreSQL Row Lock ( `SELECT FOR UPDATE` ) | **Redis Lua Scripting.** Menjalankan validasi saldo, penguncian harga, dan anti-sniping dalam satu siklus memori (< 5 ms), menjamin SLA < 300 ms bebas *lock contention*. |
| **Mekanisme Agunan Finansial** | Saldo Mengendap Wajib Pasif | **Dynamic Hold on Bid.** Saldo peserta ditahan hanya saat menawar lelang aktif, dan dicairkan 100% saat lelang berakhir jika tidak menang. |
| **Metode Penawaran Peserta** | Input Bebas Form Angka | **Discrete Increment Buttons.** Menghilangkan risiko kesalahan ketik (*typo*) dan memastikan kenaikan harga teratur sesuai ketentuan kurator. |
| **Penyimpanan Gambar Produk** | AWS S3 / Cloud Bucket | **PostgreSQL BYTEA (BLOB).** Menjaga portabilitas sistem lelang secara mandiri dan menghasilkan output Base64 terkompresi langsung via API. |
| **Arsitektur Autentikasi** | Single Auth Provider | **Dual-Login (NIRO SSO + Internal Superadmin).** Mengintegrasikan SSO eksternal organisasi untuk peserta sekaligus mempertahankan kedaulatan kontrol administratif master platform. |

## 10. Document Sign-Off Section

Dokumen arsitektur tingkat tinggi (*High-Level Design*) ini telah ditinjau secara komprehensif, diselaraskan dengan kebutuhan kompetisi Hackathon 2026 dan template resmi perusahaan, serta disahkan sebagai acuan implementasi arsitektur platform.

| PROJECT MANAGER<br>**Geri Setyo Andrian** | TECH LEAD / TECHNICAL ARCHITECT<br>**Alini** | SYSTEM ANALYST / SOLUTION ARCHITECT<br>**Daniel H. S.** |
|---|---|---|
| *[ Tanda Tangan / Signature ]* | *[ Tanda Tangan / Signature ]* | *[ Tanda Tangan / Signature ]* |
| **Tanggal:** ___ / ___ / 2026<br>**Status:** [ ] Disetujui [ ] Revisi | **Tanggal:** ___ / ___ / 2026<br>**Status:** [ ] Disetujui [ ] Revisi | **Tanggal:** ___ / ___ / 2026<br>**Status:** [ ] Disetujui [ ] Revisi |